

Greedy Dispatch – Mathematical Formulation

Gati Shakti Vishwavidyalaya

1 What Greedy Does

Greedy assigns trips to units one at a time in departure-time order. No lookahead, no correction. Single forward pass. *Code: `src/algorithms/greedy.py`, uses `greedy_construction()` from `src/utls/greedy_init.py`*

2 Assignment Rule

For each trip t sorted by dep_t (departure minute):

$$u^* = \arg \min_{u \in U_{avail}(t)} t_{loc(u), orig(t)}$$

Where $U_{avail}(t)$ = units that can reach $orig(t)$ before dep_t .

Assign trip t to unit u^* . Update unit location and availability.

3 Cost Evaluated After Assignment

$$Cost_{greedy} = w_1 D_{head} + w_2 N_{units} + w_4 P_{uncov}$$

No timing violations by construction (only feasible arcs assigned). *Computed by: `compute_cost()` in `src/utls/cost.py`*

4 Complexity

Time: $O(T \cdot N)$ where T = trips, N = units.

Space: $O(T + N)$.

Greedy MWIS complexity as per Guth Jarkovsky et al.: $O(|V|^2)$.

5 Why Greedy Fails

A locally optimal early assignment can prevent globally better assignments later. In our experiment: 756 passengers uncovered, 5 overloaded trips. This mirrors the paper’s finding: greedy lacks foresight for global solution quality.